

Ali Tiles

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Security Clearance: DoD Investigation in Progress

EDUCATION

Northeastern University

Boston, MA

Bachelor of Science in Computer Science and Sociology

Expected August 2028

- **Coursework:** Algorithms & Data Structures, Object-Oriented Design, Artificial Intelligence, Discrete Structures
- **Activities:** Video Editing Club, Arab Student Association

EXPERIENCE

MIT Lincoln Laboratory

Lexington, MA

Web Application Developer (AI Integration) Co-op

Jun 2026 – Present

- Owning design and implementation of an LLM-backend search layer over the Laboratory's Apache Solr index, authoring the technical proposal and driving it through engineering review ahead of schedule
- Engineered retrieval logic in Java that ranks and narrows candidate documents before model invocation, cutting query response time 70% and token consumption per request 40%
- Delivered a working end-to-end prototype and demoed it to the group, unifying SharePoint, internal documentation, and the public Laboratory site behind a single search interface
- Built the ingestion layer on Norconex, handling authenticated access, JavaScript-rendered pages, and mixed document formats to normalize multiple Laboratory web properties and SharePoint into the Solr index

Top Choice Realty

Staten Island, NY

Frontend Developer Intern

Jun 2025 – Sep 2025

- Architected reusable React/TypeScript component patterns to standardize dashboard and intake workflows, improving UI consistency across the app
- Refactored MongoDB schema design to enforce consistency across client and agent records, eliminating duplicate entries
- Designed and implemented Python and C# automation pipelines to synchronize distributed MongoDB records across virtualized environments, reducing manual reconciliation time by 30%

Robert DeFalco Realty

Staten Island, NY

Computer Technician Intern

Jun 2023 – Sep 2023

- Developed PowerShell automation for provisioning and deployment across 20+ systems, standardizing Windows and Linux configurations

PROJECTS

Eternal2x (DaVinci Resolve Smart Upscale) | [Website](#) | [GitHub](#)

Jan. 2026 – Present

- *Tech Stack:* Python, Lua, OpenCV, FFmpeg
- Engineered a DaVinci Resolve plugin that decodes footage through OpenCV's FFmpeg backend, computes per-frame motion scores via tile-based analysis, and groups high-motion frames into segments
- Automated marker-based cutting and timeline regrouping, applying 2x upscale with Optical Flow interpolation on motion segments and Nearest on static segments for speed
- Shipped a one-click Windows and macOS installer with an auto-updater across tagged releases

Eternal Summary | [Website](#) | [GitHub](#)

Sep. 2023 – Present

- *Tech Stack:* JavaScript, Chrome MV3, Node/Express, Docker, Fly.io, Gemini API
- Built a Chrome extension (Manifest V3) using a background service worker and injected content scripts to extract live page text and return one-click AI summaries
- Deployed a containerized Node/Express backend on Fly.io to proxy Gemini API calls, scoping host permissions to the backend origin and keeping credentials in runtime secrets

TECHNICAL SKILLS

Languages: Java, Python, TypeScript, JavaScript, C#, C++, SQL, PowerShell

Frameworks: React, Next.js, Node.js/Express, Tailwind CSS

Databases & Search: PostgreSQL, MySQL, MongoDB, Apache Solr

Infrastructure & Tools: Docker, Linux, Git/GitHub, Fly.io, Jira, Boomi, SharePoint, OpenCV, Gemini API